

802.11 CSMA/CA Wireless Link and Interference Model for the ncsim Networked Compute Simulator

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Abstract

This document describes the 802.11 CSMA/CA wireless link and interference model implemented in the `ncsim` discrete-event simulator. The model encompasses three layers: (1) a physical layer based on log-distance path loss with MCS rate adaptation from IEEE 802.11n/ac/ax tables, (2) a MAC layer incorporating Bianchi's saturation throughput analysis for CSMA/CA contention, and (3) a spatial reuse model using conflict graphs derived from carrier sensing range under the protocol interference model. Two model variants are provided: a static clique-based fair-share model (`csma_clique`) and a dynamic SINR-aware model with Bianchi MAC efficiency (`csma_bianchi`). All equations presented here correspond exactly to the software implementation.

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1 Introduction and Motivation

The `ncsim` simulator models networked computing scenarios where computational tasks are distributed across nodes and connected by communication links. In wireless deployments—particularly those using IEEE 802.11 (WiFi)—the effective throughput of each link depends on:

1. **Distance-dependent SNR:** Path loss determines the received signal quality, which selects the modulation and coding scheme (MCS) and thus the PHY data rate.
2. **CSMA/CA contention:** Links sharing the same spectrum coordinate via carrier sensing; nearby links cannot transmit simultaneously, reducing per-link throughput.
3. **Hidden terminal interference:** Links outside each other’s carrier sensing range may transmit simultaneously, causing SINR degradation at the receiver.

Prior to this model, `ncsim` used a simple proximity-based interference model (factor = $1/k$ for k nearby active links) with statically specified link bandwidths. The model described here replaces both with physically-grounded computations derived from RF parameters and IEEE 802.11 specifications.

2 RF Configuration Parameters

The model is parameterized by an immutable `RFConfig` structure with the following fields:

Table 1: RF configuration parameters and defaults.

Parameter	Symbol	Default	Description
<code>tx_power_dBm</code>	P_{tx}	20	Transmit power (dBm)
<code>freq_ghz</code>	f	5.0	Carrier frequency (GHz)
<code>path_loss_exponent</code>	n	3.0	Path loss exponent
<code>noise_floor_dBm</code>	N_0	−95	Noise floor (dBm)
<code>cca_threshold_dBm</code>	θ_{CCA}	−82	CCA detect threshold (dBm)
<code>channel_width_mhz</code>	W	20	Channel width (MHz)
<code>wifi_standard</code>	—	<code>ax</code>	MCS table: <code>n</code> , <code>ac</code> , <code>ax</code>
<code>shadow_fading_sigma</code>	σ_{SF}	0.0	Shadow fading std. dev. (dB)
<code>rts_cts</code>	—	<code>false</code>	RTS/CTS enabled

These parameters can be specified in the scenario YAML file under `config.rf` and overridden via CLI flags (`--tx-power`, `--freq`, `--path-loss-exponent`, `--wifi-standard`, `--rts-cts`).

3 Physical Layer Model

3.1 Log-Distance Path Loss

The path loss model uses the standard log-distance formulation with a Friis free-space reference at distance $d_0 = 1$ m:

$$\text{PL}(d_0) = 20 \log_{10} \left(\frac{4\pi d_0 f}{c} \right) \quad (1)$$

where f is the carrier frequency in Hz and $c = 3 \times 10^8$ m/s. At 5 GHz with $d_0 = 1$ m, this gives $\text{PL}(d_0) \approx 46.4$ dB. At 2.4 GHz, $\text{PL}(d_0) \approx 40.0$ dB.

For distances $d > d_0$, the path loss grows logarithmically:

$$\text{PL}(d) = \text{PL}(d_0) + 10 n \log_{10} \left(\frac{d}{d_0} \right) \quad (2)$$

where n is the path loss exponent ($n = 2$ for free space, $n = 3$ – 4 for typical indoor/outdoor environments).

Implementation notes:

- For $d \leq 0$: $\text{PL} = 0$ (collocated nodes, no loss).
- For $0 < d < d_0$: d is clamped to d_0 (near-field effects are not modeled).

3.2 Received Power

The received power at distance d is:

$$P_{\text{rx}}(d) = P_{\text{tx}} - \text{PL}(d) - X_{\text{SF}} \quad (3)$$

where X_{SF} is the shadow fading component in dB (see Section 3.5). All quantities are in dBm.

3.3 Signal-to-Noise Ratio (SNR)

The SNR at the receiver, in the absence of co-channel interference, is:

$$\text{SNR} = P_{\text{rx}} - N_0 \quad (\text{in dB}) \quad (4)$$

This determines the baseline PHY rate for each link.

3.4 Signal-to-Interference-plus-Noise Ratio (SINR)

When hidden terminal interferers are present, SINR replaces SNR. Interference powers must be summed in the *linear* (milliwatt) domain:

$$\text{SINR} = 10 \log_{10} \left(\frac{P_{\text{rx}}^{(\text{linear})}}{N_0^{(\text{linear})} + \sum_{j \in \mathcal{H}} P_j^{(\text{linear})}} \right) \quad (5)$$

where $\mathcal{H}$ is the set of active hidden terminal links (those *not* in the conflict graph neighborhood—see Section 7), and the conversion between dBm and milliwatts is:

$$P^{(\text{linear})} = 10^{P^{(\text{dBm})}/10} \quad (6)$$

Key insight: Only hidden terminals contribute to SINR degradation. Links within the CSMA contention domain do *not* transmit simultaneously (carrier sensing prevents it), so they affect throughput via time-sharing, not via interference power. This separation is central to the `csmabianchi` model (see Section 9.2).

3.5 Shadow Fading

When $\sigma_{\text{SF}} > 0$, log-normal shadow fading is applied. For each ordered node pair (i, j) , a fading value is drawn from:

$$X_{ij} \sim \mathcal{N}(0, \sigma_{\text{SF}}^2) \quad (7)$$

The fading map is:

- **Symmetric:** $X_{ij} = X_{ji}$ (reciprocal channel).
- **Deterministic:** Generated from a seeded PRNG (`random.Random(seed)`), ensuring identical fading across runs with the same seed.
- **Per-node-pair:** Each unique pair (i, j) with $i < j$ gets one independent draw.

When $\sigma_{\text{SF}} = 0$ (the default), the shadow fading map is empty and $X_{\text{SF}} = 0$ for all links.

4 MCS Rate Adaptation

The PHY data rate is selected from an MCS (Modulation and Coding Scheme) table based on the link SNR. The implementation includes tables for three 802.11 standards, each for 1 spatial stream at 20 MHz channel width with long guard interval:

4.1 MCS Tables

Table 2: 802.11ax (HE) MCS table, 20 MHz, 1 spatial stream, $3.2 \mu\text{s}$ GI.

MCS	Modulation	Min SNR (dB)	Rate (Mbps)
0	BPSK 1/2	5	8.6
1	QPSK 1/2	8	17.2
2	QPSK 3/4	11	25.8
3	16-QAM 1/2	14	34.4
4	16-QAM 3/4	18	51.6
5	64-QAM 2/3	22	68.8
6	64-QAM 3/4	25	77.4
7	64-QAM 5/6	29	86.0
8	256-QAM 3/4	32	103.2
9	256-QAM 5/6	35	114.7
10	1024-QAM 3/4	38	129.0
11	1024-QAM 5/6	41	143.4

4.2 Rate Selection Algorithm

Given a link SNR value, the highest MCS index whose minimum SNR requirement is met is selected:

$$R_{\text{PHY}} = \max\{r_k : \text{SNR} \geq \text{SNR}_{\text{min},k}\} \quad (8)$$

Table 3: 802.11ac (VHT) MCS table, 20 MHz, 1 SS.

MCS	Modulation	Min SNR (dB)	Rate (Mbps)
0	BPSK 1/2	5	6.5
1	QPSK 1/2	8	13.0
2	QPSK 3/4	11	19.5
3	16-QAM 1/2	14	26.0
4	16-QAM 3/4	18	39.0
5	64-QAM 2/3	22	52.0
6	64-QAM 3/4	25	58.5
7	64-QAM 5/6	29	65.0
8	256-QAM 3/4	32	78.0
9	256-QAM 5/6	35	86.7

Table 4: 802.11n (HT) MCS table, 20 MHz, 1 SS, 800 ns GI.

MCS	Modulation	Min SNR (dB)	Rate (Mbps)
0	BPSK 1/2	5	6.5
1	QPSK 1/2	8	13.0
2	QPSK 3/4	11	19.5
3	16-QAM 1/2	14	26.0
4	16-QAM 3/4	18	39.0
5	64-QAM 2/3	22	52.0
6	64-QAM 3/4	25	58.5
7	64-QAM 5/6	29	65.0

where $(r_k, \text{SNR}_{\min,k})$ are the rate and threshold for MCS index k . If the SNR is below the minimum MCS threshold (MCS 0), the rate is 0 (link not viable).

For channel widths other than 20 MHz, rates scale linearly:

$$R_{\text{PHY}}(W) = R_{\text{PHY}}(20) \times \frac{W}{20} \quad (9)$$

This is approximate but standard practice for capacity estimation.

4.3 Unit Conversion

The simulator uses MB/s (megabytes per second) internally. PHY rates are converted from Mbps:

$$R \text{ [MB/s]} = \frac{R \text{ [Mbps]}}{8} \quad (10)$$

5 Per-Link PHY Rate Computation

For each link ℓ connecting transmitter node i to receiver node j , the PHY rate is computed as follows:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (11)$$

$$P_{rx} = P_{tx} - \text{PL}(d_{ij}) - X_{ij} \quad (12)$$

$$\text{SNR}_\ell = P_{rx} - N_0 \quad (13)$$

$$R_\ell = \frac{1}{8} \text{MCS_lookup}(\text{SNR}_\ell, \text{standard}, W) \quad (14)$$

This rate R_ℓ (in MB/s) is what is assigned to `link.bandwidth` before the simulation begins.

Explicit bandwidth override: If a link specifies an explicit `bandwidth` value in the scenario YAML, that value is preserved and the RF-derived rate is not applied. This allows mixing wired and wireless links in the same scenario.

6 Carrier Sensing Range

The carrier sensing range d_{CS} is the maximum distance at which a transmission is detected by the Clear Channel Assessment (CCA) mechanism:

$$P_{tx} - \text{PL}(d_{CS}) = \theta_{CCA} \quad (15)$$

Solving for d_{CS} :

$$d_{CS} = d_0 \cdot 10^{\frac{P_{tx} - \theta_{CCA} - \text{PL}(d_0)}{10n}} \quad (16)$$

With default parameters ($P_{tx} = 20$ dBm, $\theta_{CCA} = -82$ dBm, $f = 5$ GHz, $n = 3$), the carrier sensing range is approximately 71.2 m.

Note that the carrier sensing range is determined by the CCA threshold, while the communication range is determined by the receiver sensitivity (noise floor plus minimum SNR threshold). Because the CCA threshold (-82 dBm) is higher (less sensitive) than the receiver sensitivity ($-95 + 5 = -90$ dBm), the carrier sensing range can be *shorter* than the communication range. This is a well-known property of 802.11 systems and is one source of hidden terminal problems.

7 Conflict Graph

The conflict graph $G_C = (L, E_C)$ is defined over the set of links L . Two links ℓ_a and ℓ_b are connected by an edge in E_C (i.e., they *conflict*) if carrier sensing prevents them from transmitting simultaneously. This follows the *protocol interference model*.

7.1 Conflict Definition

Let $\text{tx}(\ell)$ and $\text{rx}(\ell)$ denote the transmitter and receiver nodes of link ℓ , respectively. Let $d(u, v)$ denote the Euclidean distance between nodes u and v .

7.1.1 Without RTS/CTS

Links ℓ_a and ℓ_b conflict if either transmitter can sense any node of the other link:

$$(\ell_a, \ell_b) \in E_C \iff \exists (u, v) \in \{\text{tx}(\ell_a)\} \times \{\text{tx}(\ell_b), \text{rx}(\ell_b)\} \cup \{\text{tx}(\ell_b)\} \times \{\text{tx}(\ell_a), \text{rx}(\ell_a)\} : d(u, v) \leq d_{CS} \quad (17)$$

In other words, only *transmitters* perform carrier sensing. The receiver does not sense the channel before the transmitter initiates.

7.1.2 With RTS/CTS

When RTS/CTS is enabled, the virtual carrier sensing mechanism protects both transmitters and receivers. Any node of one link sensing any node of another link creates a conflict:

$$(\ell_a, \ell_b) \in E_C \iff \exists (u, v) \in \{\text{tx}(\ell_a), \text{rx}(\ell_a)\} \times \{\text{tx}(\ell_b), \text{rx}(\ell_b)\} : u \neq v \wedge d(u, v) \leq d_{CS} \quad (18)$$

RTS/CTS reduces hidden terminal problems by extending the conflict graph, at the cost of reduced spatial reuse.

7.2 Maximum Clique Computation

For the `csma_clique` model (Section 9.1), each link ℓ needs the size of the largest clique containing it in the conflict graph:

$$\omega(\ell) = \max\{|C| : C \text{ is a clique in } G_C \text{ and } \ell \in C\} \quad (19)$$

The implementation uses:

- **Bron-Kerbosch with pivoting** (exact) for networks with ≤ 50 links.
- **Greedy approximation** for larger networks: for each link ℓ , greedily grow a clique from ℓ by adding neighbors that are pairwise connected.

8 Bianchi's CSMA/CA MAC Efficiency Model

The MAC-layer efficiency of IEEE 802.11 DCF is modeled using Bianchi's saturation throughput analysis [1]. This gives the fraction of channel time used for successful payload delivery when n stations contend.

8.1 Coupled Equations

Bianchi's model solves two coupled nonlinear equations for the transmission probability τ and the conditional collision probability p :

$$\tau = \frac{2(1 - 2p)}{(1 - 2p)(W_{\min} + 1) + pW_{\min}(1 - (2p)^m)} \quad (20)$$

$$p = 1 - (1 - \tau)^{n-1} \quad (21)$$

where:

- $W_{\min} = 16$ is the minimum contention window (CW_{min} for 802.11),
- $m = 6$ is the maximum backoff stage (CW_{max} = $W_{\min} \cdot 2^m = 1024$),
- n is the number of contending stations.

These equations are solved by iterative fixed-point iteration, initialized with $\tau = 2/(W_{\min} + 1)$ and iterated until convergence ($|\tau_{\text{new}} - \tau| < 10^{-10}$, max 200 iterations).

8.2 Throughput Efficiency

From the converged τ , three probabilities characterize each virtual time slot:

$$P_{\text{idle}} = (1 - \tau)^n \quad (22)$$

$$P_{\text{success}} = n \tau (1 - \tau)^{n-1} \quad (23)$$

$$P_{\text{collision}} = 1 - P_{\text{idle}} - P_{\text{success}} \quad (24)$$

The expected duration of a virtual slot is:

$$\mathbb{E}[T_{\text{slot}}] = P_{\text{idle}} \cdot T_{\text{idle}} + P_{\text{success}} \cdot T_{\text{success}} + P_{\text{collision}} \cdot T_{\text{collision}} \quad (25)$$

with the following slot durations (5 GHz OFDM PHY):

Table 5: Bianchi model slot duration parameters.

Parameter	Description	Value (μs)
T_{idle}	Empty slot duration	9
T_{success}	Successful frame (DATA + SIFS + ACK)	500
$T_{\text{collision}}$	Collision duration	500

The MAC efficiency (fraction of time carrying successful payload) is:

$$\eta(n) = \frac{P_{\text{success}} \cdot T_{\text{success}}}{\mathbb{E}[T_{\text{slot}}]} \quad (26)$$

Properties:

- $\eta(1) = 1.0$ (single station, no contention).
- $\eta(n)$ is monotonically decreasing in n .
- $\eta(n) > 0$ for all $n \geq 1$.
- Typical values: $\eta(2) \approx 0.88$, $\eta(5) \approx 0.71$, $\eta(10) \approx 0.59$.

The implementation precomputes $\eta(n)$ for $n = 1, \dots, 100$ in a lazy-initialized lookup table.

9 Interference Model Variants

The `ncsim` execution engine computes the effective bandwidth for each link as:

$$B_{\text{eff}}(\ell) = \frac{B_{\ell} \cdot f(\ell)}{N_{\ell}} \quad (27)$$

where B_{ℓ} is the link's base bandwidth (`link.bandwidth`), $f(\ell) \in (0, 1]$ is the interference factor, and N_{ℓ} is the number of concurrent flows sharing link ℓ . The two WiFi model variants differ in how B_{ℓ} and $f(\ell)$ are set.

9.1 Variant 1: Static Clique Model (`csma_clique`)

The static clique model computes all contention effects at setup time and bakes them into the link bandwidth:

$$B_\ell = \frac{R_\ell}{\omega(\ell)} \quad (28)$$

where R_ℓ is the SNR-derived PHY rate (from Section 5) and $\omega(\ell)$ is the maximum clique size containing ℓ (from Section 7.2).

The interference factor is always 1.0:

$$f_{\text{clique}}(\ell) = 1.0 \quad (29)$$

Rationale: In the worst case, all links in the maximum clique are simultaneously active, and CSMA/CA time-sharing gives each an equal fraction. The clique bound provides a conservative upper bound on the number of mutually contending links. This is a static, worst-case analysis independent of which links are actually active.

Properties:

- No dynamic recalculation needed (`get_affected_links` returns $\emptyset$).
- Conservative: may underestimate throughput when not all clique members are active.
- Computationally cheap at runtime.

9.2 Variant 2: Dynamic Bianchi Model (`csma_bianchi`)

The dynamic model correctly separates two distinct interference mechanisms:

1. **Contention domain** (conflict graph neighbors): CSMA prevents simultaneous transmission. These links share airtime via Bianchi’s model.
2. **Hidden terminals** (active links *not* in the conflict graph): These may transmit simultaneously, causing SINR degradation at the receiver.

9.2.1 Setup

The link bandwidth is set to the SNR-only PHY rate (no contention reduction):

$$B_\ell = R_\ell \quad (30)$$

9.2.2 Dynamic Interference Factor

Let $\mathcal{A}$ be the set of currently active links. For link ℓ , define:

$$\mathcal{C}_\ell = \mathcal{A} \cap \mathcal{N}(\ell) \quad (\text{active contending links—conflict graph neighbors}) \quad (31)$$

$$\mathcal{H}_\ell = (\mathcal{A} \setminus \mathcal{N}(\ell)) \setminus \{\ell\} \quad (\text{active hidden terminals—not in conflict graph}) \quad (32)$$

where $\mathcal{N}(\ell)$ is the set of conflict graph neighbors of ℓ .

SINR factor (hidden terminals). If $\mathcal{H}_\ell \neq \emptyset$, the SINR at the receiver of ℓ is computed per Equation (5), with interference from hidden terminal transmitters only. The degraded rate is looked up from the MCS table:

$$f_{\text{SINR}}(\ell) = \frac{R_{\text{SINR}}(\ell)}{R_\ell} \quad (33)$$

where $R_{\text{SINR}}(\ell) = \text{MCS_lookup}(\text{SINR}_\ell)$ and R_ℓ is the base SNR-only rate. If no hidden terminals are active, $f_{\text{SINR}} = 1.0$. If the SINR drops below the minimum MCS threshold, $f_{\text{SINR}} = 0.01$ (link effectively destroyed).

Contention factor (Bianchi). The number of contending stations is $n = 1 + |\mathcal{C}_\ell|$ (the link itself plus its active contending neighbors). Each station gets a fraction $\eta(n)/n$ of the channel:

$$f_{\text{contention}}(\ell) = \frac{\eta(n)}{n} \quad (34)$$

Combined factor.

$$f_{\text{bianchi}}(\ell) = f_{\text{SINR}}(\ell) \cdot f_{\text{contention}}(\ell) = \frac{R_{\text{SINR}}(\ell)}{R_\ell} \cdot \frac{\eta(n)}{n} \quad (35)$$

clamped to $[0.01, 1.0]$.

9.2.3 Dynamic Recalculation

When a transfer starts or completes on a link, all active conflict graph neighbors are recalculated:

$$\text{get_affected_links}(\ell) = \mathcal{N}(\ell) \cap \mathcal{A} \quad (36)$$

This triggers re-evaluation of the interference factor and rescheduling of completion events for affected transfers.

9.3 Comparison of Variants

Table 6: Comparison of WiFi interference model variants.

Property	csma_clique	csma_bianchi
Link bandwidth setup	$R_\ell/\omega(\ell)$	R_ℓ
Interference factor	1.0 (static)	$\frac{R_{\text{SINR}}}{R_\ell} \cdot \frac{\eta(n)}{n}$ (dynamic)
Contention model	Worst-case clique bound	Bianchi MAC + active count
Hidden terminals	Not modeled	SINR degradation
Recalculation trigger	None	Active neighbor changes
Runtime cost	$O(1)$ per event	$O(\mathcal{N}(\ell))$ per event

10 Integration with the Execution Engine

The WiFi model integrates with `ncsim` via two existing abstractions, requiring *zero changes* to the core execution engine:

1. `link.bandwidth` is overwritten at setup time with the WiFi-derived PHY rate (or PHY rate / clique size for `csma_clique`).
2. `InterferenceModel.get_interference_factor()` returns the dynamic factor $f(\ell)$, which the engine multiplies with the link bandwidth before applying per-link fair sharing:

Listing 1: Engine bandwidth computation (`execution_engine.py`).

```

1 def _get_link_bandwidth(self, link_id, ...):
2     link = self.network.links[link_id]
3     if self.interference_model is None:
4         return link.bandwidth
5     active_links = self._get_active_link_ids()
6     factor = self.interference_model.get_interference_factor(
7         link_id, active_links, self.network
8     )
9     return link.bandwidth * factor

```

The per-flow effective bandwidth is then:

$$B_{\text{per_flow}}(\ell) = \frac{B_\ell \cdot f(\ell)}{N_\ell} \quad (37)$$

Transfer completion time is:

$$t_{\text{complete}} = t_{\text{now}} + \frac{D_{\text{remaining}}}{B_{\text{per_flow}}} + L_{\text{total}} \quad (38)$$

where $D_{\text{remaining}}$ is the remaining data in MB and L_{total} is the total link latency.

11 Worked Example

Consider the `wifi_test.yaml` scenario with default RF parameters:

- Four nodes: $n_0(0, 0)$, $n_1(30, 0)$, $n_2(0, 30)$, $n_3(30, 30)$.
- Two links: $\ell_{01} : n_0 \rightarrow n_1$ and $\ell_{23} : n_2 \rightarrow n_3$.
- Each link carries a 50 MB transfer.

11.1 PHY Rate Computation

For link ℓ_{01} with $d = 30$ m:

$$\begin{aligned}
\text{PL}(d_0 = 1\text{m}) &= 20 \log_{10} \left(\frac{4\pi \cdot 1 \cdot 5 \times 10^9}{3 \times 10^8} \right) \approx 46.4 \text{ dB} \\
\text{PL}(30\text{m}) &= 46.4 + 10 \cdot 3 \cdot \log_{10}(30) \approx 46.4 + 44.3 = 90.7 \text{ dB} \\
P_{\text{rx}} &= 20 - 90.7 = -70.7 \text{ dBm} \\
\text{SNR} &= -70.7 - (-95) = 24.3 \text{ dB} \\
\text{MCS lookup} &\Rightarrow \text{MCS 5 (64-QAM 2/3), min SNR} = 22 \text{ dB} : 68.8 \text{ Mbps} \\
R_{\ell_{01}} &= 68.8/8 = 8.6 \text{ MB/s}
\end{aligned}$$

Both links have identical geometry, so $R_{\ell_{23}} = 8.6 \text{ MB/s}$.

11.2 Conflict Graph

Carrier sensing range: $d_{\text{CS}} = 71.2 \text{ m}$. The transmitter n_0 is at distance 30 m from n_2 and 42.4 m from n_3 ; both within d_{CS} . Therefore ℓ_{01} and ℓ_{23} conflict. $\omega(\ell_{01}) = \omega(\ell_{23}) = 2$.

11.3 csma_clique Result

$$\begin{aligned}
B_{\ell_{01}} &= 8.6/2 = 4.3 \text{ MB/s}, \quad f = 1.0 \\
t_{\text{transfer}} &= 50/4.3 = 11.63\text{s} \\
\text{Makespan} &= 0.01\text{s (task)} + 11.63\text{s (transfer)} + 0.01\text{s (task)} = 11.65\text{s}
\end{aligned}$$

11.4 csma_bianchi Result

$$\begin{aligned}
B_{\ell_{01}} &= 8.6 \text{ MB/s (base rate)} \\
n &= 2 \text{ contending stations} \\
\eta(2) &\approx 0.88, \quad f_{\text{contention}} = 0.88/2 = 0.44 \\
f_{\text{SINR}} &= 1.0 \text{ (no hidden terminals—both links are in conflict graph)} \\
f &= 1.0 \times 0.44 = 0.44 \\
B_{\text{eff}} &= 8.6 \times 0.44 = 3.78 \text{ MB/s} \\
t_{\text{transfer}} &= 50/3.78 = 13.21\text{s} \\
\text{Makespan} &\approx 13.22\text{s}
\end{aligned}$$

The Bianchi model gives a longer transfer time because $\eta(2)/2 \approx 0.44 < 1/2 = 0.50$ (the clique model's share). The difference accounts for MAC protocol overhead (backoff slots, collisions) that the clique model's simple $1/\omega$ bound does not capture.

12 YAML Configuration

Listing 2: WiFi scenario YAML example.

```

1 scenario:
2   name: "WiFi CSMA Bianchi Test"
3   network:

```

```

4     nodes:
5         - {id: n0, compute_capacity: 1000, position: {x: 0, y: 0}}
6         - {id: n1, compute_capacity: 1000, position: {x: 30, y: 0}}
7     links:
8         # No bandwidth -- derived from RF model
9         - {id: l01, from: n0, to: n1, latency: 0.0}
10        # Explicit bandwidth preserved as override
11        - {id: l23, from: n2, to: n3, bandwidth: 50, latency: 0.0}
12    config:
13        interference: csma_bianchi    # or csma_clique
14        rf:
15            tx_power_dBm: 20
16            freq_ghz: 5.0
17            path_loss_exponent: 3.0
18            noise_floor_dBm: -95
19            cca_threshold_dBm: -82
20            channel_width_mhz: 20
21            wifi_standard: "ax"
22            shadow_fading_sigma: 0.0
23            rts_cts: false

```

13 Summary

The 802.11 WiFi model in `ncsim` provides a physically-grounded replacement for simple heuristic interference models. The key contributions are:

1. **Distance-derived link rates** from log-distance path loss and 802.11n/ac/ax MCS tables, eliminating the need to manually specify wireless link bandwidths.
2. **Protocol-model conflict graph** from carrier sensing range, with optional RTS/CTS for receiver protection.
3. **Two complementary interference variants:** a fast static clique bound and a dynamic Bianchi-based model that captures both CSMA contention and hidden terminal SINR degradation.
4. **Zero engine modifications:** the model composes cleanly with `ncsim`'s existing `link.bandwidth` and `InterferenceModel` abstractions.

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